

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>

#include <sys/ipc.h>

#include <sys/shm.h>

#include <string.h>


int main() {

    int pipefd[2];

    pid_t pid;

    char pipe_message[] = "Hello from parent through pipe!";

    char buffer[100];


    // ----- PIPE IPC -----

    if (pipe(pipefd) == -1) {

        perror("Pipe creation failed");

        exit(1);

    }


    pid = fork();


    if (pid < 0) {

        perror("Fork failed");

        exit(1);

    }
```

```
if (pid > 0) {  
    // Parent process  
    close(pipefd[0]);  
  
    write(pipefd[1], pipe_message, strlen(pipe_message) + 1);  
  
    close(pipefd[1]);  
    wait(NULL);  
}  
else {  
    // Child process  
    close(pipefd[1]);  
  
    read(pipefd[0], buffer, sizeof(buffer));  
  
    printf("PIPE IPC:\n");  
    printf("Child received: %s\n\n", buffer);  
  
    close(pipefd[0]);  
    exit(0);  
}  
  
// ----- SHARED MEMORY IPC -----  
  
int shmid;
```

```
char *shared_memory;
```

```
shmid = shmget(IPC_PRIVATE, 1024, IPC_CREAT | 0666);
```

```
if (shmid < 0) {
```

```
    perror("Shared memory creation failed");
```

```
    exit(1);
```

```
}
```

```
pid = fork();
```

```
if (pid < 0) {
```

```
    perror("Fork failed");
```

```
    exit(1);
```

```
}
```

```
if (pid > 0) {
```

```
    // Parent process
```

```
    shared_memory = (char *)shmat(shmid, NULL, 0);
```

```
    if (shared_memory == (char *)-1) {
```

```
        perror("Shared memory attachment failed");
```

```
        exit(1);
```

```
    }
```

```
    strcpy(shared_memory, "Hello from parent through shared memory!");
```

```
printf("SHARED MEMORY IPC:\n");  
printf("Parent wrote: %s\n", shared_memory);  
  
wait(NULL);  
  
shmdt(shared_memory);  
shmctl(shmid, IPC_RMID, NULL);  
}  
else {  
    // Child process  
    shared_memory = (char *)shmat(shmid, NULL, 0);  
  
    if (shared_memory == (char *)-1) {  
        perror("Shared memory attachment failed");  
        exit(1);  
    }  
  
    sleep(1);  
  
    printf("Child read: %s\n", shared_memory);  
  
    shmdt(shared_memory);  
    exit(0);  
}
```

```
return 0;
```

```
}
```